



# Electrical, optical and morphological properties of Au/n-Ge heterostructures by using cobalt phthalocyanine (CoPc) interlayer

Pavani M<sup>a</sup>, Ashok Kumar A<sup>a,\*</sup>, Rajagopal Reddy V<sup>b</sup>, Kaleemulla S<sup>c</sup>, Jyothi I<sup>d</sup>, Chel-Jong Choi<sup>d</sup>

<sup>a</sup> Department of Physics, YSR Engineering College of Yogi Vemana University, Proddatur, YSR Kadapa (Dt), Andhra Pradesh, India

<sup>b</sup> Department of Physics, Sri Venkateswara University, Tirupati, Andhra Pradesh, India

<sup>c</sup> Thin Films Laboratory, Centre for Functional Materials, Vellore Institute of Technology, Vellore, Tamil Nadu, India

<sup>d</sup> Semiconductor Physics Research Center (SPRC), School of Semiconductor and Chemical Engineering, Jeonbuk National University, Jeonju 54896, Republic of Korea

## ARTICLE INFO

### Keywords:

n-Ge  
CoPc  
Trap charge limited current  
Electrical properties  
Metal-Phthalocyanine  
Photodiode properties

## ABSTRACT

The morphology of CoPc thin films on n-Ge substrate characterized using AFM measurements shows fairly smooth surface with RMS roughness of 8.42 nm. Structural analysis of cobalt phthalocyanine (CoPc) thin films is confirmed using Raman measurements at 785 nm excitation wavelength with three vibrational modes at 1539  $\text{cm}^{-1}$  resembles the C-N bond and 755  $\text{cm}^{-1}$ , 685  $\text{cm}^{-1}$  resembles to C-C bonds. The EDS spectrum of the Au/CoPc/n-Ge surface reveals the presence of elements contributing the structure of CoPc. Optical properties of CoPc thin films described with a Q band splitting between 600 to 700 nm with an optical bandgap of 3.8 eV. The electrical measurements of Au/n-Ge Schottky heterostructures with CoPc interlayer reveals that the barrier height parameters are significantly improved than without interlayer. Low series resistance values of heterostructures than pristine contact shows its applicability in efficient photovoltaic cells. The conduction mechanisms of these contacts are evaluated using forward current-voltage properties. At moderately high forward voltages, the transport properties of Au/CoPc/n-Ge heterostructures are truly dominated by trap charge limited current (TCLC) mechanism. The photodiode properties of CoPc/n-Ge heterostructures were investigated at 775 nm wavelength and the responsivity and detectivity of the contacts found to be varied with reverse voltage.

## 1. Introduction

As the silicon technology reached to its size limits in the fabrication of high speed CMOS devices, a very suitable semiconductor replaces the silicon is germanium. During the past decade, germanium semiconductor plays a very important role in the fabrication of high mobility MOS devices due to its high electron and hole mobility than compared to its silicon counterparts [1,2]. A very common problems encountered with Ge-based devices are Fermi level pinning near valence band edge and thermal stability at high temperatures. The electrical properties of Ge-based devices are significantly unaffected with the Fermi level pinning even for different metal contacts [3]. The challenges mentioned above in the case of Ge-based devices can be alleviated by inserting a thin interlayer between metal-Ge semiconductors. A thin interlayer may be an organic semiconductor, high-k dielectric oxide material. A thin insulating layer inserted in between MS junctions converts it into a metal/insulator/semiconductor (MIS) device. The Schottky barrier height (SBH) of the junctions can be manipulated by the formation of a

dipole layer with the help of a thin oxide or organic materials between the metal and semiconductor. Several initiatives have been taken to alter the barrier height of heterojunctions by utilizing an organic semiconductor, an insulating layer, and/or a chemical passivation method [4]. The metal induced gap states (MIGS) model was successfully explained by Nishimura to understand the Fermi level pinning (FLP) at metal-Ge interface [5]. To alleviate FLP, two approaches were proposed such as inserting ultra-thin interfacial layer and choosing a low free-electron metal as a Schottky contact metal. Understanding of metal/Ge interface plays a significant role in controlling the band alignment at the metal/Ge interface which helps to design high speed Ge n-channel devices. Ultra-low resistance (Tb) inserted in between Ni-Ge/p-Ge contacts shows enhanced SBH due to accumulation of Tb atoms at the interface [6].

A thin layer between metal-semiconductor junctions includes polymers, ceramics, high-k insulators or oxide layers which modify the semiconductor surface and that can reduce the MIGS and density of interface states. It has also been observed that the thin layers of polymer

\* Corresponding author.

E-mail address: [drashok.yvuce@gmail.com](mailto:drashok.yvuce@gmail.com) (A.K. A.).

<https://doi.org/10.1016/j.mtcomm.2023.106360>

Received 9 April 2023; Received in revised form 27 May 2023; Accepted 1 June 2023

Available online 2 June 2023

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materials (CoPc) improve the quality of rectification properties of heterojunction than compared to contacts without interlayer [7]. Organic semiconductors with p-type conductivity have been studied widely due to their applications in electronic and optoelectronic devices. Metal-phthalocyanine (MPc) materials are classification of organic substances which are found to be thermally stable and also stable against ambient conditions. A few classifications of MPc materials can be used in the fabrication of OLEDs [8], OPV cells, OFETs and organic spintronic devices as functional interfaces to metals. Cobalt phthalocyanine (CoPc) is one of the members of the MPc family which shows the p-type conductivity [9]. The conductivity and ease of fabrication of CoPc organic semiconductor found its suitability in various applications such as biosensors and humidity sensors.

Pt/Ge heterojunctions were realized using a low-cost carbon paste (CP) as an interlayer and reviewed the electrical conduction properties [10]. These properties reveal good rectification properties, high barrier height than contact without interlayer and highly stable electrical conduction properties. Rawat et al. [11] investigated the negative magneto-conductance or magneto-resistance properties of cobalt phthalocyanine (CoPc) thin films and analysed its suitability for organic or molecular spintronic applications. The negative magneto-capacitance value was found to be decreased with the applied bias which is possibly due to trap-assisted bipolaron. Tataroglu et al. [12] demonstrated the fabrication of Au/CoPc:GO/p-Si/Al heterojunction diodes at different concentrations of graphene oxide (GO) using spin coating technique and investigated the electrical characteristics by the transient photocurrent, photovoltage and photoconductance measurements. This heterojunction exhibits photoconductive properties under illumination showing its application as a photodiode. Ahmet et al. and Baraz et al. [13,14] fabricated Au/n-Si (MS) and metal-polymer-semiconductor (MPS) Schottky contacts using the derivatives of CoPc thin layer using spin coating procedure and studied their current transport properties. They observed a profound influence of organic interlayer (CoPc and CuPc derivatives) on the electrical properties of the Schottky contacts. In addition, they noted that the presence of organic semiconductor layer between metal-semiconductor junctions not only affects the change of interface states distribution but also reduces leakage current while enhancing the magnitude of current in the forward bias zone. Wahab et al. [9] studied the electrical measurements of CoPc/n-Si heterostructures at different illumination intensities revealing the CoPc/n-Si behaves as a photosensor. Wahab et al. [15] also investigated the transport mechanism of CoPc/p-Si heterojunction which exhibits good rectification characteristics with a significant rectification ratio. Popielarski et al. [16] studied the electrical properties of n-Si/metal phthalocyanine (MPc) interfaces. The metal phthalocyanine ( $M = \text{Co}, \text{Cu}$ ) powders are vacuum coated on Si surface and heat treated the contacts from 293 to 500 K. Sekhar Reddy et al. was investigated the surface topography, optical and electrical properties of Au/CuPc/n-Si heterojunction at different CuPc film thickness [17]. It was observed that the electrical properties of the Au/n-Si Schottky junction can be significantly altered with the variation of CuPc thickness as an interlayer. Further, increasing the thickness of the CuPc thin film on n-Si reduces the defect density and the localized states. Very recently, the authors studied the electrical properties of n-Ge contacts modified by CoPc interlayer using I-V measurements but a detailed investigation correlating the surface topography, structural analysis of CoPc thin films and distribution of interface states at CoPc/Ge interface is very much essential [7].

In the current investigation, n-Ge heterostructure were realized with CoPc as an interlayer and studied its current-voltage (I-V) in the dark and illumination and capacitance-voltage (C-V) measurements. In the case of Schottky devices, the current transport behavior plays a significant role in understanding the electrical properties of the CoPc/n-Ge contacts. Moreover, the observed enhancement of barrier properties of CoPc/n-Ge heterostructure such as reverse leakage currents, barrier height and series resistance show its suitability in various optoelectronic

and photovoltaic device applications. Photodiode property of Au/CoPc/n-Ge heterostructure shows an enhanced photocurrent in both reverse and forward voltages at an excitation wavelength of 775 nm.

## 2. Materials and methods

In the current study, an n-type Ge wafer (100) (Sb-doped) wafer with (100) orientation was employed. Firstly, the Ge-wafer was cleaned with trichloroethylene, acetone and methanol using ultrasonic agitation. Then the n-Ge samples were etched in  $\text{H}_2\text{O}:\text{HF}$  (100:1) solution. After each stage of cleaning, the wafer was rinsed in deionized water with a resistivity of 15  $\text{M}\Omega\text{-cm}$ . Subsequently, the wafer was dried in  $\text{N}_2$  atmosphere and immediately transferred to a vacuum coating unit to deposit low resistance ohmic contact (Al) of 50 nm thickness on the rough side of the sample. A thin CoPc organic layer (purchased from Tokyo chemical industries co. Ltd. with CAS NO. 3317-67-7 having a purity of >93%) is vacuum deposited using the thermal evaporation method on the polished side of the Ge wafer. After the formation of the thin CoPc layer, Schottky metal (Au) was evaporated using electron beam evaporation with a shadow mask of 0.5 mm diameter. The thickness of Au was read as 40 nm using a digital thickness monitor. All vacuum evaporation of metals and CoPc polymer were carried out at a base pressure of  $1 \times 10^{-6}$  Torr. A pristine sample was fabricated using the similar process without CoPc interlayer. Morphology of CoPc thin films on cleaned n-Ge surfaces were investigated using AFM and FESEM measurements. Structural properties of CoPc thin films on glass plate were analysed using Raman spectroscopy at an excitation wavelength of 785 nm. Electrical measurements of the pristine and heterostructure contacts were performed at room temperature using I-V source meter (Agilent 4155 C) and an LCR meter (Agilent 4284 A). The photodetection measurements of Au/CoPc/n-Ge heterostructure were carried out in the presence of illumination of light at 775 nm wavelength and technical details of the measurement was found in earlier reports [18].

## 3. Results and discussion

### 3.1. Morphological, structural and optical properties of CoPc thin films

The surface morphology of the heterostructures plays an important role in understanding the electrical properties of the device. In the present study, AFM and FESEM measurements were performed for CoPc thin films. AFM topography (2D & 3D images) of the CoPc thin films is presented in Fig. 1 with  $4 \times 4 \mu\text{m}^2$  scan area. The topography reveals grains and grain boundaries with a relatively smooth surface having RMS roughness of 8.42 nm. FESEM image and EDS spectrum of the heterostructures are shown in Fig. 2(a) and (b) respectively. It is observed from the FESEM image (Fig. 2(a)) that the spherical grains are distributed uniformly on the entire surface of the sample. The EDS spectrum of Au/CoPc/n-Ge heterostructure shown in Fig. 2(b) represents with atomic percentages of different elements. The EDS spectrum of the heterostructure shows the signals of Ge, C, Au, Co, N and O with atomic percentages mentioned at the inset of Fig. 2(b). It is observed the presence of CoPc chemical composition ( $\text{C}_{32}\text{H}_{16}\text{CoN}_8$ ) in the EDS analysis at low weight percentages might be due to the very thin CoPc interfacial layer. Very low concentration of Co atoms observed in EDS spectra could be due to its chemical stoichiometric composition. The cross-sectional FESEM image of Au/CoPc/n-Ge heterostructures was presented in the Fig. 3(a). The micrograph reveals a sharp interface between CoPc and Ge layers and no specific interfacial reactions were noticed. The average thickness of the CoPc interlayer was found to be  $\sim 70$  nm. The at% of CoPc elements also depends on its thickness. As the thickness of the CoPc is of the order of 70 nm, the at% of the CoPc elements seems to be in low magnitudes than usual on par with its stoichiometric composition.

The dense cracks are observed in the FESEM surface morphology of Au/CoPc/n-Ge heterostructure. According to Cranston and Lessard [19],

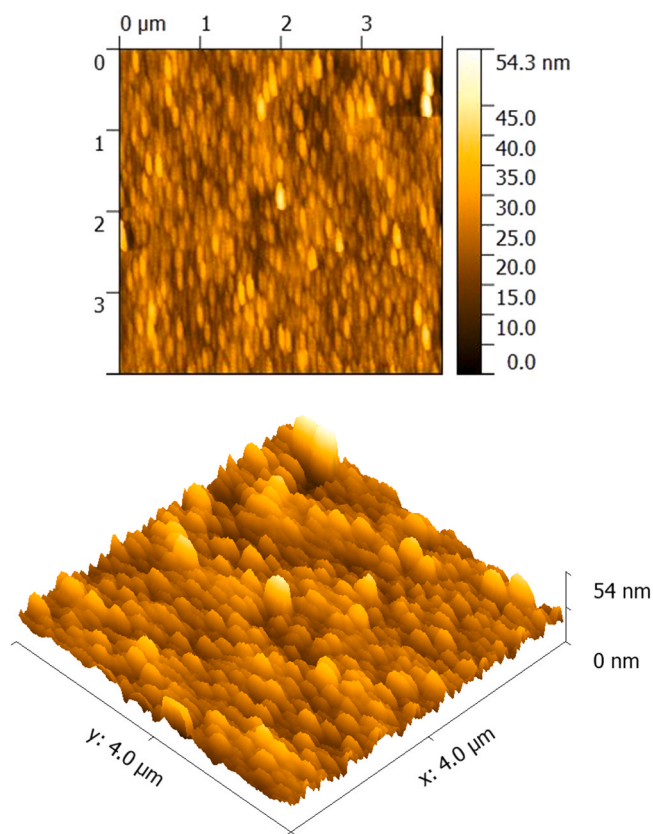


Fig. 1. Atomic force microscopy (AFM) image of the CoPc thin films.

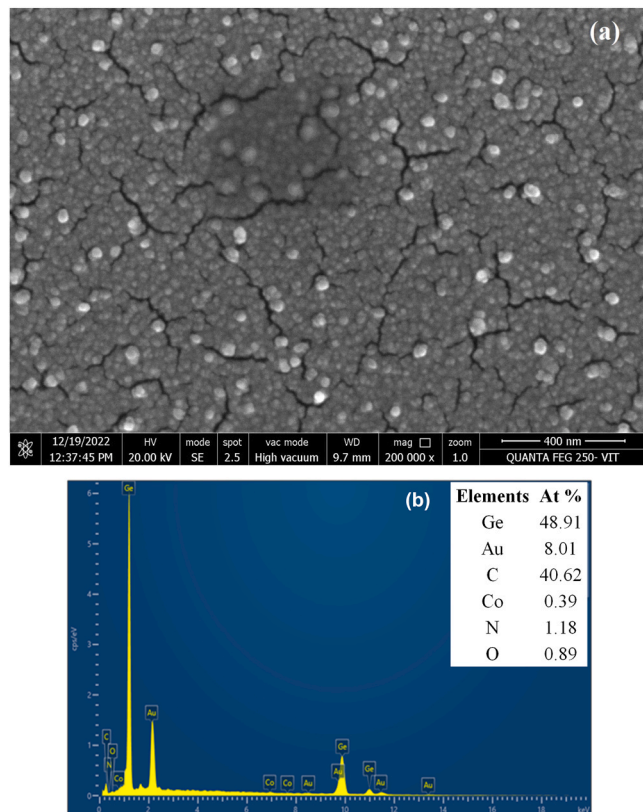


Fig. 2. (a) FESEM image and (b) EDX spectra of Au/CoPc/n-Ge heterostructure.

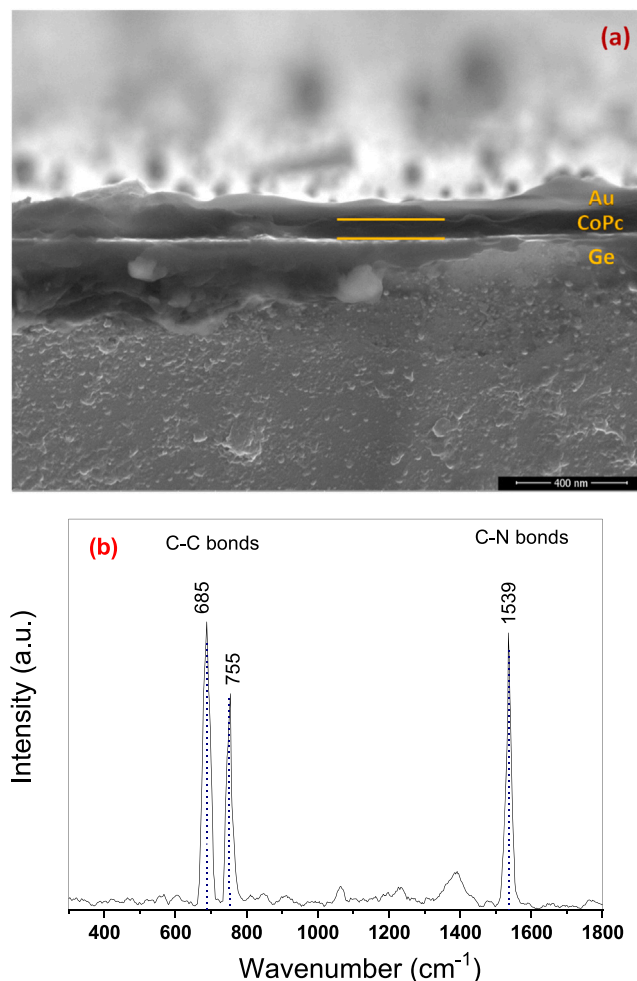


Fig. 3. (a) Cross-sectional FESEM image of Au/CoPc/n-Ge heterostructure and (b) Raman measurements of thin CoPc layer on glass substrate.

the thermodynamic growth of organic thin films especially metal-phthalocyanines (MPCs) rely with different parameters such as substrate temperature and deposition rates particularly in the case of physical vapour deposition methods. The deposition rates considerably influence the nucleation density which defines the growth of thin film formation. The density of molecules reaching the substrate surface increases under large deposition rates pertaining to form smaller and denser crystallites. On the other hand, the substrate (n-Ge) surface roughness also influences the growth of thin film and the nucleation process. A large surface roughness increases the density of nucleation centres which further decreases the diffusion distance of evaporated molecules. This generally causes to form small grains distribution, increased voids or discontinuities and also leads to form distinct molecular orientation. In the present work, the observed voids or discontinuities on the surface morphology might be associated with the abnormality in deposition rates causes to form a layered growth of MPCs on the Ge-substrate. Other reason could be associated with the irregularities of the roughness of the n-Ge substrate which might form organic thin films on the substrate with small grains and increased voids.

In the present work, dense cracks in nanometer dimension are observed in the surface morphology. It is also feasible to accept that the Schottky metal contact at the crack may diffuse inside and favorably alters the current conduction properties. But due to the layered growth of the CoPc thin film, the dense cracks could probably be on the surface and is not penetrated deeply causing the metal contact not having intimate contact with the n-Ge surface. This could also be justified by seeing the FESEM cross section as shown in Fig. 3(a). But still a detailed



investigation is very much necessary to analyse the cause in the formation of discontinued surface morphology of organic MPCs.

The structural property of thermally evaporated CoPc organic layer on glass substrates was determined using Raman measurements at an excitation wavelength of 785 nm in the spectral range from 300  $\text{cm}^{-1}$  to 1800  $\text{cm}^{-1}$ . The Raman spectra showed in Fig. 3(b) reveal three significant vibrations at 1539  $\text{cm}^{-1}$ , 755  $\text{cm}^{-1}$  and 685  $\text{cm}^{-1}$ . Molecular vibrations occur at 1545  $\text{cm}^{-1}$  might be associated with the C-N bonds whereas at 755  $\text{cm}^{-1}$  and 685  $\text{cm}^{-1}$  might be due to C-C bonds. The vibrational modes observed in the present investigation are well comparable with the earlier work implying the confirmation of CoPc structure [20,21].

Optical absorption properties of CoPc thin films were also employed using UV-VIS measurements in the wavelength range from 300 to 900 nm as shown in Fig. 4. Different metal-phthalocyanine compounds shows optical transition from 1.5 eV to 6.2 eV which are termed with different bands such as Q, B, N, L and C. Normally Q bands appears in the visible region from 600 nm to 700 nm, while the other bands appear in the UV region. A significant Q band splitting is observed (from the inset of Fig. 4(a)) with two significant peaks (labelled as Q<sub>1</sub> and Q<sub>2</sub>) in the optical spectra of CoPc thin films at 618 nm and 690 nm and found similar with the earlier reports [22–25]. These broadened Q bands observed in the visible region of the spectra are associated from  $\pi$ - $\pi^*$  transitions and is generally attributed to ring structure of metal-phthalocyanines. The energy separation (Davydov splitting) between Q<sub>1</sub> (2.0 eV) and Q<sub>2</sub> (1.78 eV) which is 0.22 eV is well matched

with the literature [25]. The optical bandgap is measured from Tauc's plot usually drawn between  $(\alpha h\nu)^2$  versus photon energy is shown in Fig. 4(b). The bandgap of the CoPc thin films were found to be 3.8 eV is nearly matched with the reports published [23,25].

### 3.2. Electrical properties of Au/n-Ge heterostructures

Fig. 5(a) depicts the I-V measurements of Au/n-Ge Schottky contact and Au/CoPc/n-Ge heterostructure at room temperature. The reverse leakage current of the heterostructure was found to be smaller than that of the Au/n-Ge Schottky contact. For the bias voltages  $V > 3kT/q$ , considering the series resistance ( $R_s$ ) of Schottky contacts and interfacial layer, the net current due to thermionic emission theory might be expressed as [14],

$$I = I_0 \left[ \exp \left( \frac{q(V - IR_s)}{nkT} \right) - 1 \right] \left\{ 1 - \exp \left[ \frac{q(V - IR_s)}{kT} \right] \right\} \quad (1)$$

where the parameters mentioned have usual meaning and  $A^{**}$  was assumed as 140 for n-Ge [26].  $I_0$  is the saturation current obtained from the intercept of  $\ln(I)$  versus  $V$  plot at  $V = 0$  and is expressed as

$$I_0 = AA^{**}T^2 \exp \left( \frac{q\Phi_{B0}}{kT} \right) \quad (2)$$

The current-voltage characteristics follows a linear relation at very low voltages and the slope of this linear region is evaluated as ideality factor ( $n$ ) which quantifies the current transport due to thermionic emission. This can be expressed as

$$n = \frac{q}{kT} \frac{dV}{d(\ln I)} \quad (3)$$

The barrier height and ideality factor values are evaluated as 0.63 eV and 1.51 for Au/n-Ge contact and 0.67 eV and 1.58 for the Au/CoPc/n-Ge heterostructure, respectively. The ideality factor is found to be much greater than unity in both contacts. This could be associated with barrier inhomogeneity where electrons conduct through low barrier height patches [27]. The derivation of ideality factor greater than unity in the case of Au/n-Ge with CoPc interlayer might also be associated with the presence of CoPc organic layer of non-uniform thickness, series resistance, generation and recombination of carriers through interface states, presence of dipoles due to the organic interlayer and also due to metal-induced defects at the interface [6,28]. The carrier transport in the Schottky contacts with semiconductor substrates having high doping concentration generally seems to deviate from the ideal thermionic emission [29–31]. The high barrier height of Au/n-Ge contacts observed in the present work than compared to earlier work might be due to the current transport mechanism dominated by metal induced gap states or surface states presented as the metal-Ge interface [32,33]. The barrier height of the Au/CoPc/n-Ge heterostructure was found to be greater than that of the Au/n-Ge contact. An interfacial layer (CoPc) between metal-Ge Schottky contact preferably used to modulate the SBH by influencing the space charge region. This CoPc interfacial layer also causes to depinning of Fermi level at n-Ge semiconductor surface which further lowers the tunneling resistance [34]. The CoPc interfacial layer also causes to shift the work function of the metal and semiconductor electron affinity which also may be the reason for the enhanced barrier height. CoPc interfacial layer also acts as a physical barrier between metal-Ge interfaces. Similar type of results were also presented recently in the study of electrical properties of Au/Carbon paste/n-Ge contacts and Au/Graphene/n-Ge contacts [10,35]. These studies show that the interfacial layer significantly enhances the electrical properties by varying the band alignment.

To evaluate the effective barrier height, we have tested 20 diodes of Au/n-Ge and Au/CoPc/n-Ge contacts. The barrier heights of the contacts were evaluated in the forward bias using thermionic emission theory at linear voltages range from 0.01 V to 0.1 V. The statistical distribution of

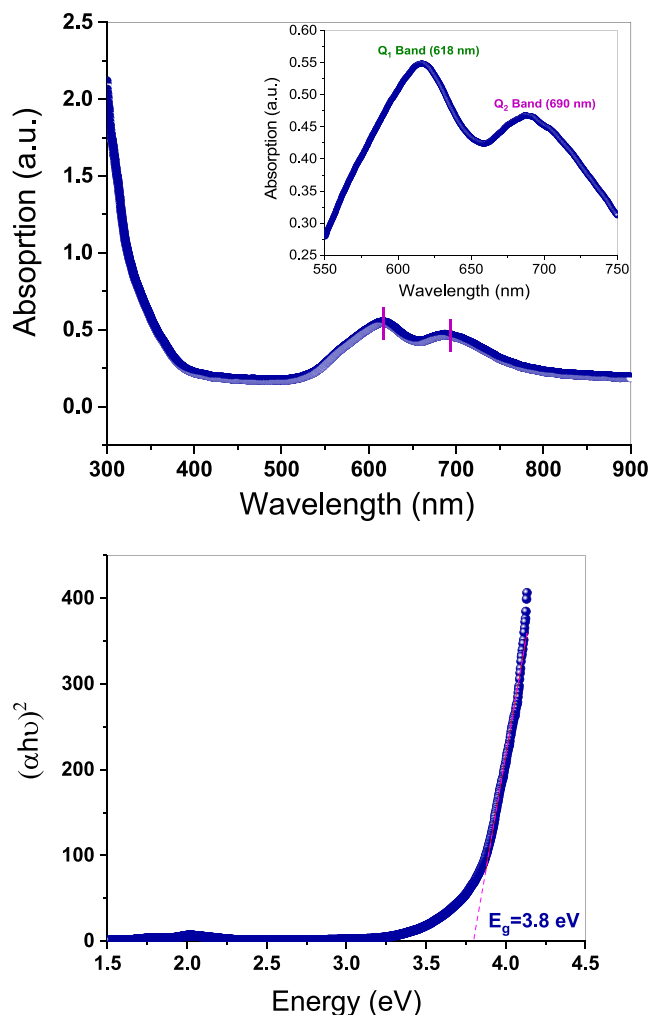


Fig. 4. (a) Optical absorption properties of CoPc thin films on glass plate (b) Tauc's plot to determine the optical band gap of the CoPc thin films.

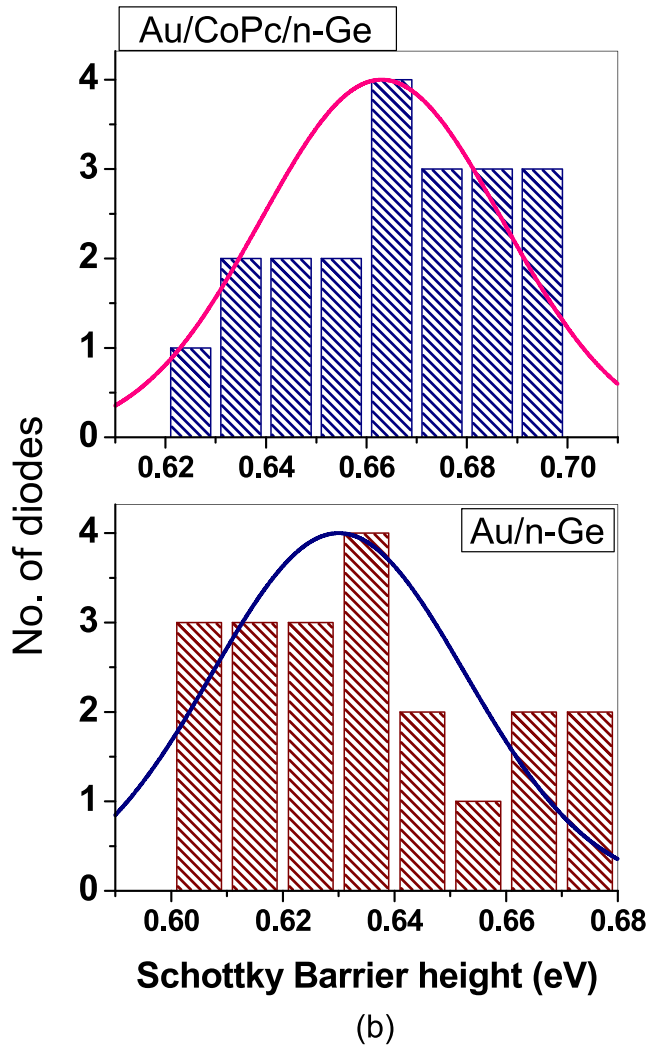
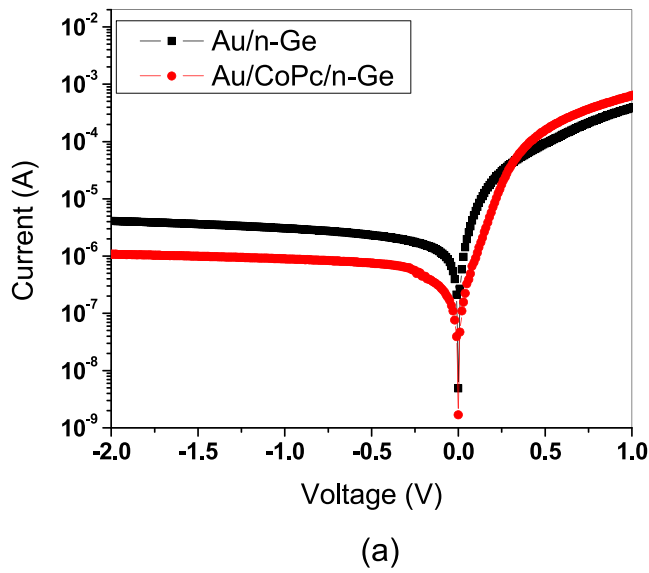


Fig. 5. (a) Current-voltage properties of Au/n-Ge Schottky contact and Au/CoPc/n-Ge heterostructure (b) Statistical distribution of barrier height of 20 diodes of Au/n-Ge and Au/CoPc/n-Ge Schottky contacts.

barrier heights of Au/n-Ge and Au/CoPc/n-Ge contacts was shown in Fig. 5(b). The barrier heights of Au/n-Ge contacts varied from 0.60 to 0.68 and the Gaussian distribution yields a mean barrier height of 0.63 eV with standard deviation of 13 meV. On the other hand, the barrier height of the Au/CoPc/n-Ge contacts shows a variation from 0.62 eV to 0.70 eV with mean barrier height of 0.67 eV and standard deviation of 15 meV.

The current-voltage characteristic of the contacts deviates from linearity at sufficiently large voltages due to the presence of series resistance irrespective of interface states density at equilibrium with the semiconductor. Generally, Cheung method can be used to evaluate the series resistance ( $R_s$ ) in the non-linear region of the forward bias I-V measurements [36]. By considering the  $R_s$ , the Cheung's functions are described below,

$$\frac{dV}{d(\ln I)} = \frac{nkT}{q} + IR_s \quad (4)$$

$$H(I) = V - \frac{nkT}{q} \ln \left( \frac{I}{AA^{**}T^2} \right) \quad (5)$$

$$H(I) = n\Phi_b + IR_s \quad (6)$$

Fig. 6 illustrates the plots of  $H(I)$  versus  $I$  and  $dV/d(\ln I)$  versus  $I$  for Au/n-Ge Schottky contact and Au/CoPc/n-Ge heterostructure. From the plot of  $dV/d(\ln I)$  versus  $I$  (Fig. 6(a)), the obtained ideality factor and series resistance values are 1.97 and 3.027 k $\Omega$  for Au/n-Ge Schottky contact and 2.34 and 981  $\Omega$  for Au/CoPc/n-Ge heterostructure respectively. Fig. 6(b) shows the plot of  $H(I)$  versus  $I$  and the obtained barrier

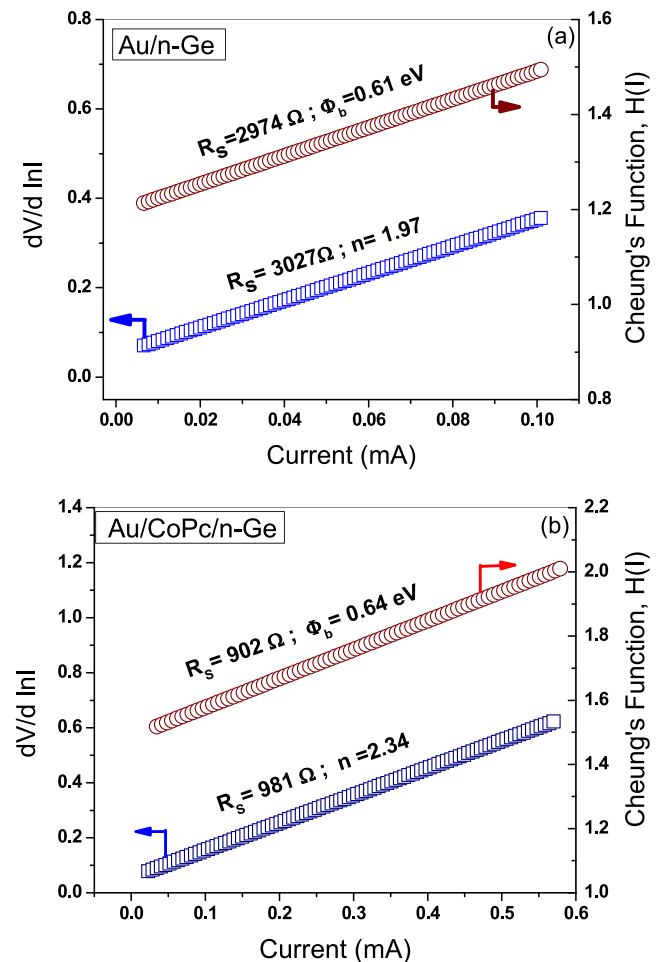


Fig. 6. (a) Cheung's plot for the Au/n-Ge contact and (b) Au/CoPc/n-Ge heterostructure.

height and series resistance values are found to be 0.61 eV and 2.974 k $\Omega$  for the Au/n-Ge Schottky contact and 0.64 eV and 902  $\Omega$  for the Au/CoPc/n-Ge heterostructure respectively. The series resistance estimated from the plots of  $dV/d(\ln I)$  versus  $I$  and  $H(I)$  versus  $I$  were closely matched with one another, reveals the validity of Cheung's function. The ideality factor obtained from the plot of  $dV/d(\ln I)$  versus  $I$  which is slightly larger than the value calculated from the linear I-V plot signifies the influence of series resistance on Schottky barrier parameters. The influence of  $R_s$  and the bias dependent BH in the non-linear portion of the forward I-V measurements can be accredited to the potential drop across the interfacial layer and modified interface states with bias [37, 38].

Norde formulated an empirical formula to determine the SBH and  $R_s$  of a Schottky contact. The following function is described using modified Norde method to determine the Schottky contact parameters [39].

$$F(V) = \frac{V}{\gamma} - \frac{kT}{q} \ln \left( \frac{I(V)}{AA^{**}T^2} \right) \quad (7)$$

where  $I(V)$  is the current and  $\gamma$  is an integer (dimensionless) which is greater than the value of  $n$  estimated from the forward I-V measurements. It is evident from the Norde plot shown in Fig. 7 that the lowest knee point of the Norde function ( $F(V_0)$ ) in the curve is used to obtain the minimum voltage ( $V_0$ ) applied across the device. The barrier height of the device, which is related to the lowest value of the Norde function, can be estimated with the following equation,

$$\Phi_{bo} = F(V_0) + \frac{V_0}{\gamma} - \frac{kT}{q} \quad (8)$$

The series resistance value can be described with the following relation,

$$R_s = \frac{kT(\gamma - n)}{qI_0} \quad (9)$$

where  $I_0$  is the current across the device for the corresponding minimum voltage  $V_0$ . Fig. 7 illustrates the plot of Norde function  $F(V)$  versus  $V$  of the Au/n-Ge Schottky contacts with and without CoPc interlayer. From the Norde plot, the obtained barrier height and  $R_s$  values are found to be 0.63 eV and 2.458 k $\Omega$  for the Au/n-Ge Schottky contact and 0.73 eV and 87.4  $\Omega$  for the Au/CoPc/n-Ge heterostructure respectively. The Schottky barrier parameters evaluated using different methods found consistence in magnitude on comparing with other methods. This implies the validity of the measurement of  $\Phi_b$ ,  $R_s$ , and  $n$  values by assuming the

thermionic emission model. However, a small discrepancy in Schottky barrier parameters derived from different approaches is observed. The barrier parameters extracted by different methods utilize the different regions of the forward I-V data for example forward  $\ln I$ -V graph is considered a linear region at low voltages, Cheung's method is applied to a non-linear region of I-V data at moderately high voltages and Norde method is employed to extract the barrier parameters in the entire region of forward I-V data [35].

Fig. 8 represents the  $1/C^2$  versus  $V$  plot of Au/n-Ge Schottky contact and Au/CoPc/n-Ge heterostructure measured at a frequency of 1 MHz. The relation between the capacitance and applied voltage can be determined as [35],

$$\frac{1}{C^2} = \left( \frac{2}{\epsilon_s q N_d A^2} \right) \left( V_{bi} - \frac{kT}{q} - V \right) \quad (10)$$

where  $\epsilon_s$  is the permittivity of the semiconductor, the x-axis intercept of  $1/C^2$  versus  $V$  plot gives  $V_0$ , built-in potential ( $V_{bi}$ ) can be derived as  $V_{bi} = V_0 + kT/q$  and the barrier height ( $\Phi_{bo}$ ) can be written as  $\Phi_{bo} = V_0 + V_n + kT/q$ , where  $V_n = (kT/q) \ln(N_c/N_d)$ . With the help of the plot, the doping concentration ( $N_d$ ) and barrier height ( $\Phi_{bo}$ ) are evaluated as  $1.65 \times 10^{16} \text{ cm}^{-3}$  and 0.73 eV for Au/n-Ge contact and  $2.26 \times 10^{16} \text{ cm}^{-3}$  and 0.86 eV for the Au/CoPc/n-Ge heterostructure respectively. The  $\Phi_{bo}$  values evaluated from the C-V measurements often found higher in magnitude than that of the I-V measurements. The capacitance of the junction is insensible to potential fluctuations on the length scale less than the space charge region; hence C-V measurements always provide mean barrier height. On the other hand, current across the junction relies exponentially on barrier height and it is also very sensitive to how surface states have distributed at the interface [40].

Charge conduction mechanisms of the Au/n-Ge Schottky contact and Au/CoPc/n-Ge heterostructure were evaluated in forward bias I-V characteristics using Log I-Log V plot. Log I-Log V plot shows different linear regions and exhibits power law behavior of current ( $I \propto V^m$ ). The magnitude of the exponent value or slope factor 'm' signifies different conduction mechanisms. Fig. 9 shows Log I-Log V plot of the Au/CoPc/n-Ge heterostructure with three linear regions having different slope factors. At low voltages ( $0.01 < V < 0.1$ ), the slope factor ( $m \sim 1.28$ ) reveals ohmic conduction is the dominant conduction mechanism which is generally caused by a reduction in the injection of carriers from electrodes. For moderately high voltages ( $0.1 < V < 0.3$ ), current and voltage parameters are related by power law dependence with a slope factor of 3.69. As the slope factor observed in this region is greater than 2, the current is truly associated with the space charge limited current (SCLC)

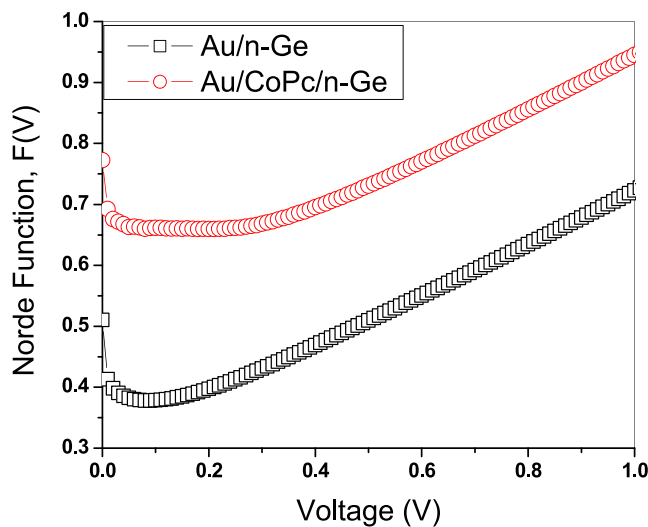


Fig. 7. Norde Plot of the Au/n-Ge Schottky contact and Au/CoPc/n-Ge heterostructure.

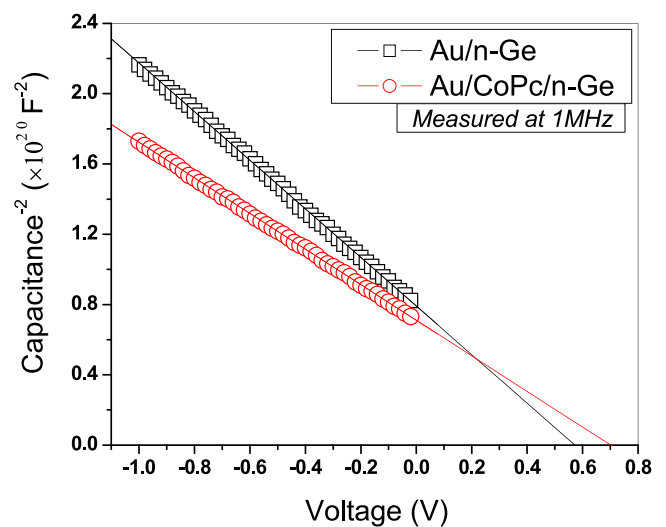


Fig. 8.  $1/C^2$  versus  $V$  graph of the Au/n-Ge and heterostructure contacts measured at 1 MHz frequency.

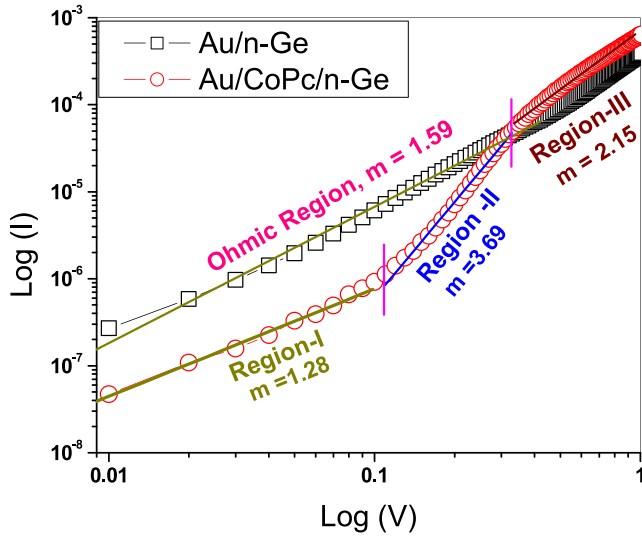


Fig. 9. Log I-Log V plot for the Au/n-Ge and heterostructure contacts.

mechanism controlled by the presence of traps distributed exponentially across the junction [41]. In this region, at high bias voltage, injection of charge carriers dominates the doping charge carriers or thermally generated carriers. This increases the density of injected carriers filling the traps which further increases the space charge of the junction. The slope factors obtained in the present work are in correlation with the earlier reports [14]. At much higher voltages ( $0.3 \text{ V} < V < 1 \text{ V}$ ) of the Au/CoPc/n-Ge heterostructure, the slope factor decreased to a value of 2 indicating the trap-free SCLC conduction mechanism. Due to the reason of strong electron injection, the electrons escape from the traps and contribute to the space-charge-limited current (SCLC) [42]. On the other hand, Au/n-Ge contact shows only one region even for a large voltage range with slope factor  $m \sim 1.59$  indicating ohmic conduction is the dominant conduction mechanism where the thermally generated carriers or doping charge carriers dominate injected carriers from the electrode.

Moreover, the electrical properties of Au/n-Ge Schottky contact and Au/CoPc/n-Ge heterostructure can be departing from ideality due to the density of interface states, voltage-dependent ideality factor  $n(V)$ , effective barrier height ( $\Phi_b$ ) and the interfacial layer. Card and Rhoderick [43] proposed the energy distribution of interface states ( $N_{ss}$ ) that can be determined as,

$$N_{ss}(V) = \frac{1}{q} \left[ \frac{\epsilon_i}{\delta} (n(V) - 1) - \frac{\epsilon_s}{W_d} \right] \quad (11)$$

where voltage-dependent ideality factor is represented as,

$$n(V) = \frac{qV}{kT \ln(I/I_0)} \quad (12)$$

where,  $\epsilon_i$  and  $\epsilon_s$  are the permittivity of the interfacial layer and semiconductor,  $\delta$  is the thickness of the insulating layer and  $W_d$  is the depletion width that can be extracted from the high frequency (1 MHz) C-V measurements. For n-type semiconductors, the energy of interface states ( $E_{ss}$ ) concerning the bottom of the conduction band ( $E_c$ ) at the semiconductor surface can be expressed as,

$$E_c - E_{ss} = q(\Phi_e - V) \quad (13)$$

where,

$$\Phi_e = \Phi_{bo} + \beta V = \Phi_{bo} + \left(1 - \frac{1}{n(V)}\right)V \quad (14)$$

Fig. 10 shows the variation of interface state density ( $N_{ss}$ ) as a

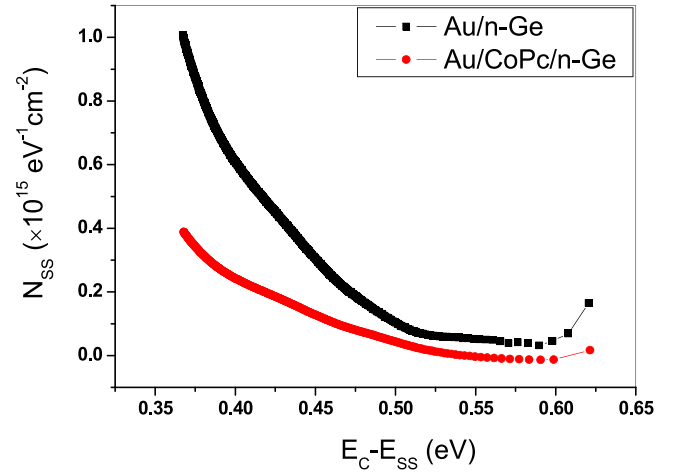


Fig. 10.  $N_{ss}$  versus  $E_c - E_{ss}$  curves of the Au/n-Ge Schottky contact and Au/CoPc/n-Ge heterostructure.

function of surface state energy ( $E_{ss}$ ) below the conduction band for the Au/n-Ge Schottky contact and Au/CoPc/n-Ge heterostructure. The density of states varied from  $1.649 \times 10^{14} \text{ eV}^{-1} \text{ cm}^{-2}$  at ( $E_c - 0.62 \text{ eV}$ ) to  $1.006 \times 10^{15} \text{ eV}^{-1} \text{ cm}^{-2}$  ( $E_c - 0.37 \text{ eV}$ ) for Au/n-Ge Schottky contacts. On the other hand, the  $N_{ss}$  values ranged from  $1.68 \times 10^{13} \text{ eV}^{-1} \text{ cm}^{-2}$  at ( $E_c - 0.62 \text{ eV}$ ) to  $3.875 \times 10^{14} \text{ eV}^{-1} \text{ cm}^{-2}$  at ( $E_c - 0.37 \text{ eV}$ ) in the case of Au/CoPc/n-Ge heterostructure. In both the contacts, the interface state density was found to increase from the middle of the band gap to the bottom of the conduction band exponentially. At surface state energy ( $E_c - 0.37 \text{ eV}$ ), the heterostructure contact shows a reduced density of states approximately by 3 orders than the Au/n-Ge Schottky contact. Similarly, at ( $E_c - 0.62 \text{ eV}$ ), the  $N_{ss}$  values of heterostructure were reduced by the factor of 10 compared to pristine contact. This clearly shows that the CoPc organic interlayer has a significant influence on the passivation of interface states even though the interlayer is unreactive and abrupt [44].

Fig. 11 (a) shows the I-V behavior of Au/CoPc/n-Ge contact under dark and illumination of light at 775 nm wavelength in the NIR region. Compared to dark I-V responses of Au/CoPc/n-Ge heterostructure, improved photocurrent is obtained in both forward and reverse bias characteristics at an irradiation of light. To understand the operating procedure of CoPc/n-Ge device, it is very much useful to interpret with a band energy diagram. The band energy levels of CoPc and n-Ge semiconductor before contact are shown in Fig. 11 (b). Under thermal equilibrium, the electrons diffuse from n-Ge to CoPc semiconductor which reduces the Fermi level position in n-Ge and increases the Fermi level in CoPc until the Fermi level matches. After an intimate contact between n-Ge and CoPc semiconductor, a depletion layer with built-in electric field is created at the interface. The exposure of light generates electron-hole pairs which are transported across the junction under the influence of electric field. The width of the depletion layer is quantitatively defined based on the bias voltage. During the forward voltage, the carriers transported easily across the CoPc/Ge junction due to lowered depletion width which significantly improves the photocurrent. On the other hand, the reverse voltage applied to the CoPc/Ge heterojunction increases the depletion width which further decreases the photocurrent [45]. The photodiode parameters such as responsivity and detectivity were evaluated using dark current ( $I_d$ ), photo current ( $I_{ph}$ ) and measured optical power density value is  $0.922 \mu\text{W}/\text{cm}^2$  at  $\lambda = 775 \text{ nm}$ . Fig. 12.

The responsivity and detectivity of the Au/CoPc/n-Ge device during reverse bias is found to be increased with bias potential which might be due to increased rate of carrier diffusion and reduction in the probability of electron-hole recombination rate at larger electric field [18]. The responsivity of the CoPc/n-Ge contact was found to be  $0.9702 \text{ A/W}$  at  $-1 \text{ V}$  and  $0.51627$  at  $-0.2 \text{ V}$ . These results signify the suitability of

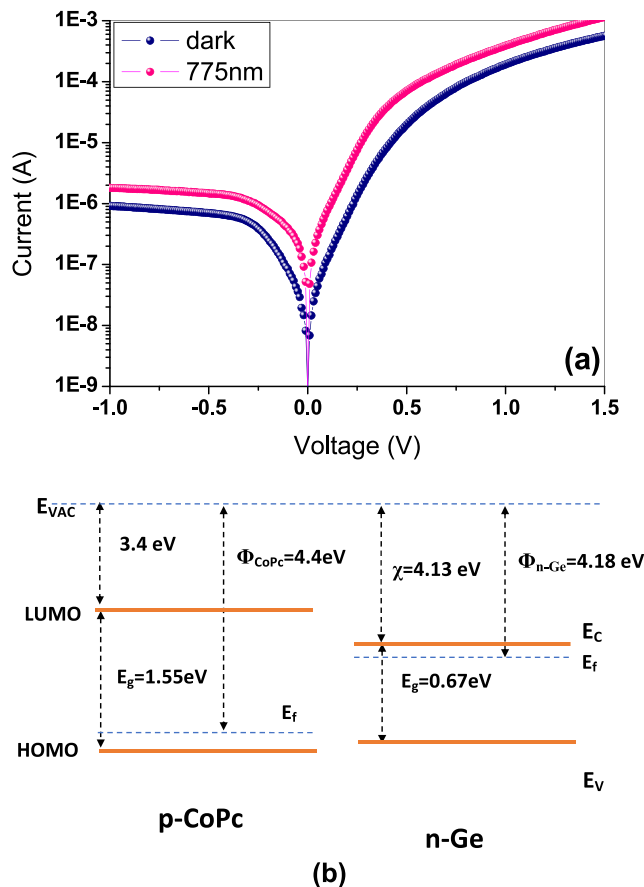


Fig. 11. (a) Photodetection behavior of Au/CoPc/n-Ge heterostructures at an illumination of 775 nm wavelength of light (b) Band energy diagram of CoPc and n-Ge semiconductor before contact.

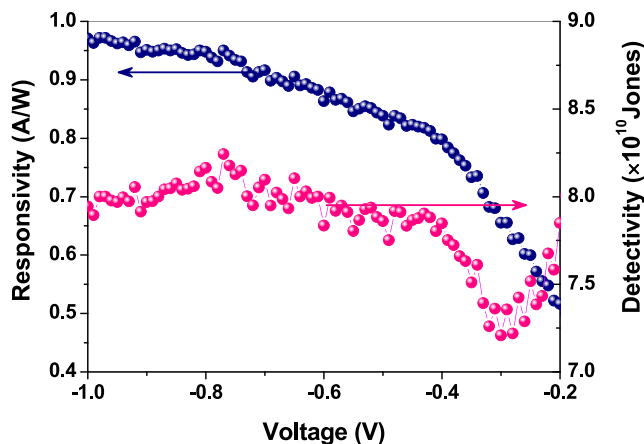


Fig. 12. Variation of responsivity and detectivity of Au/CoPc/n-Ge device for reverse voltages at an illumination of 775 nm wavelength of light.

CoPc/n-Ge heterostructure for photodiode applications. On comparing with the recent reports [46], large responsivity and small detectivity magnitudes at  $-1$  V are observed in the present work which might be due to large dark currents. Improvement in the present work is very much essential to enhance the photodetection properties of the CoPc/n-Ge heterostructures by varying the device fabrication conditions like varying the substrate temperature, modification of surface morphology of CoPc thin films and increasing the mobility of charge carriers in CoPc semiconductors usually done by doping.

#### 4. Conclusions

The effect of the CoPc interlayer on the I-V and C-V characteristics of the Au/n-Ge Schottky contact and Au/CoPc/n-Ge heterostructure were explored. Both the contacts with and without interlayer shows efficient rectification properties. Specifically high rectification ratio is observed for heterostructure contacts. Superior electrical properties of heterostructures such as larger barrier height and larger ideality factor deviating from unity might be due to reduction in the tunneling resistance and modulation of space charge across the interface. For the Au/n-Ge Schottky contact the Schottky barrier height and ideality factor values are 0.63 eV (I-V), 0.73 eV (C-V) and 1.51 and for the Au/CoPc/n-Ge Schottky contacts the SBH and ideality values are 0.67 eV (I-V), 0.86 eV (C-V) and 1.58. The Barrier height, ideality factor and series resistance for the Au/n-Ge Schottky contact and Au/CoPc/n-Ge heterostructure were determined by using Norde and Cheung's methods. The interface state density ( $N_{ss}$ ) of Au/CoPc/n-Ge Schottky contact is found to be smaller than that of the Au/n-Ge heterostructure and found that the distribution of surface states increases exponentially from the midgap. The current conduction mechanism was evaluated for the contacts using logarithmic forward I-V data. The Au/CoPc/n-Ge exhibited three different linear regions with a large slope factor ( $\sim 3.69$ ) observed in moderately high voltages. The larger slope factors can be attributed to the formation of defects or traps located at the active CoPc layer below the n-Ge conduction band. Due to this larger slope factors ( $\sim 3.69$ ), TCLC conduction mechanism dominates at moderately high voltage and current conduction is primarily through traps associated at the CoPc-Ge interface. Finally, the Au/CoPc/n-Ge heterostructure is fabricated with low series resistance, high barrier height and low reverse leakage current shows its suitability in various optoelectronic and photovoltaic applications. Under the illumination of light at 775 nm wavelength of light, the photodiode properties of Au/CoPc/n-Ge contact shows enhanced magnitudes of photocurrent in both forward and reverse biased conditions. Responsivity and detectivity values of the photodiode follows the same trend of increased magnitudes with reverse voltage.

#### Ethics Statement

I, the Corresponding Author, declare that this manuscript is original, but part of the work was published by the same authors in Materials Today: Proceedings, 2023 (In Press). <https://doi.org/10.1016/j.matpr.2023.02.097> and is being considered for publication. We herewith submit to the editorial office that we tried to include some more characterisations to correlate the findings reported earlier. I can confirm that the manuscript has been read and approved by all named authors and that there are no other persons who satisfied the criteria for authorship but are not listed. I further confirm that the order of authors listed in the manuscript has been approved by all of us. I understand that the Corresponding Author is the sole contact for the Editorial process and is responsible for communicating with the other authors about progress, submissions of revisions and final approval of proofs. Signed by the Corresponding Author on behalf of the all other authors.

#### Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

#### Data availability

Data will be made available on request.



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